

CS486/686: Introduction to Artificial Intelligence

Lecture 7c - Inference and Sampling

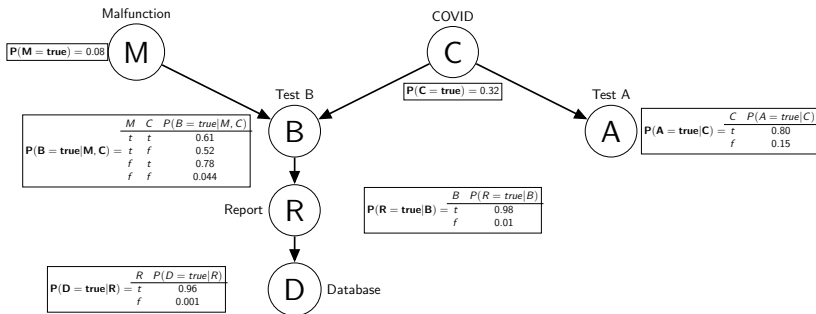
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Readings: Poole & Mackworth Chap. 9.6-9.7

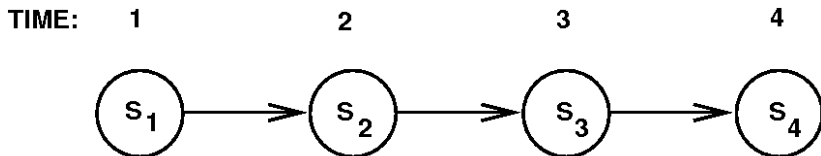
Variable Elimination Example



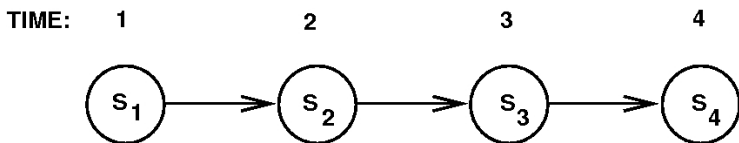
We are interested in calculating $P(C \mid D = \text{true})$
 (Problem 2 query 2 in variable-elimination.pdf, Page 10)

Probability and Time

- A node may repeat over **time**
- **Explicit** encoding of time
- Chain has length = amount of time you want to model
- **Event-driven** times or **clock-driven** times
- e.g. **Markov chain**



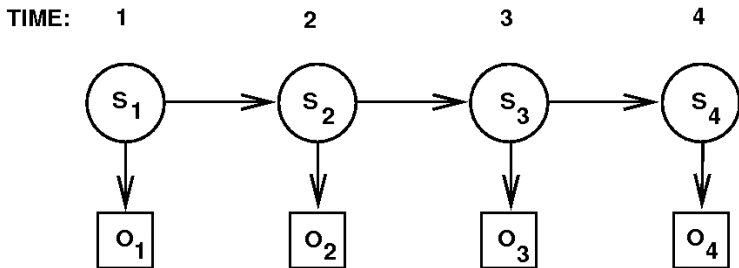
Markov Assumption



$$P(S_{t+1} \mid S_1, \dots, S_t) = P(S_{t+1} \mid S_t)$$

This distribution gives the **dynamics** of the Markov chain

Hidden Markov Models (HMMs)



Add: observations O_t (always observed, so the node is square) and observation function $P(O_t | S_t)$

Given a sequence of observations $O_1, \dots, O_t$, we can estimate:

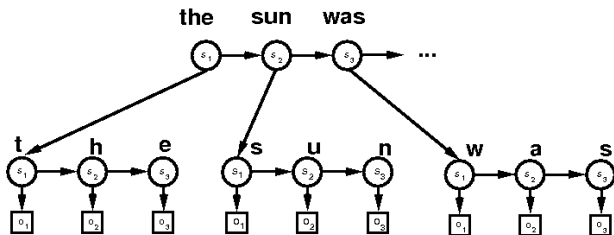
$$P(S_t | O_1, \dots, O_t)$$

or

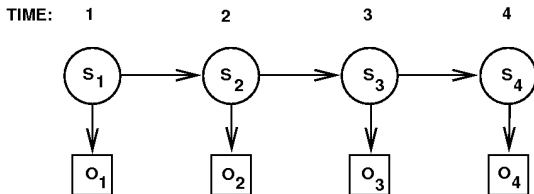
$$P(S_k | O_1, \dots, O_t) (k < t)$$

Speech Recognition

- Most well known application of HMMs
- **Observations**: audio features
- **States**: phonemes
- **Dynamics**: models e.g. co-articulation
- **HMMs**: words
- Can build hierarchical models (e.g. sentences)



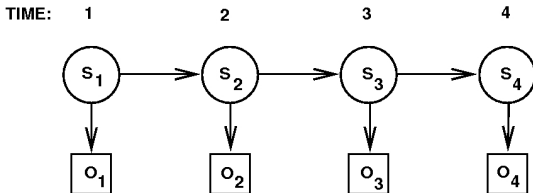
Belief Monitoring in HMMs



forward:

$$\begin{aligned}\alpha_t(i) &= P(o_{1\dots t}, S_t = i) = \sum_{s_{t-1}} P(o_{1\dots t}, s_{t-1}, S_t = i) \\ &= \sum_{s_{t-1}} P(o_t | S_t = i, o_{1\dots t-1}, s_{t-1}) P(S_t = i | o_{1\dots t-1}, s_{t-1}) P(o_{1\dots t-1}, s_{t-1}) \\ &= \sum_{s_{t-1}} P(o_t | S_t = i) P(S_t = i | s_{t-1}) P(o_{1\dots t-1}, s_{t-1}) \quad (\text{cond. indep.}) \\ &= P(o_t | S_t = i) \sum_{s_{t-1}} P(S_t = i | s_{t-1}) \alpha_{t-1}(s_{t-1})\end{aligned}$$

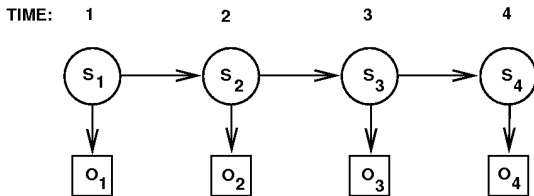
Belief Monitoring in HMMs



backward:

$$\begin{aligned}\beta_t(i) &= P(o_{t+1:T} \mid S_t = i) = \sum_{s_{t+1}} P(o_{t+1:T}, s_{t+1} \mid S_t = i) \\&= \sum_{s_{t+1}} P(o_{t+2:T} \mid o_{t+1}, s_{t+1}, S_t = i) P(o_{t+1} \mid s_{t+1}, S_t = i) P(s_{t+1} \mid S_t = i) \\&= \sum_{s_{t+1}} P(o_{t+2:T} \mid s_{t+1}) P(o_{t+1} \mid s_{t+1}) P(s_{t+1} \mid S_t = i) \quad (\text{cond. indep.}) \\&= \sum_{s_{t+1}} \beta_{t+1}(s_{t+1}) P(o_{t+1} \mid s_{t+1}) P(s_{t+1} \mid S_t = i)\end{aligned}$$

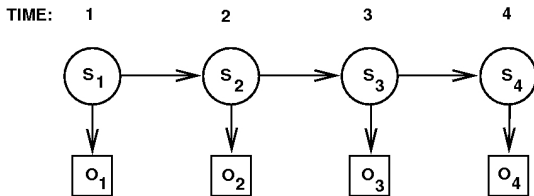
Belief Monitoring in HMMs



forward-backward:

$$\begin{aligned}\alpha_t(i)\beta_t(i) &= P(o_{1\dots t}, S_t = i)P(o_{t+1\dots T} \mid S_t = i) \\ &= P(o_{1\dots T}, S_t = i) \quad (\text{chain rule + cond. indep.})\end{aligned}$$

Belief Monitoring in HMMs

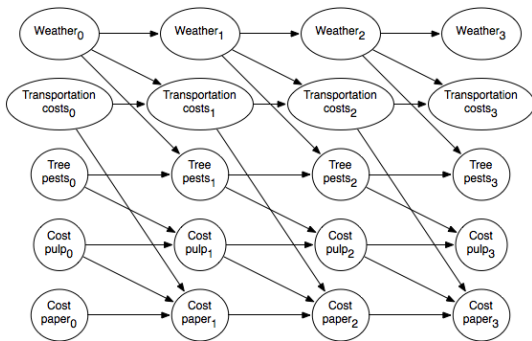


forward-backward:

$$\begin{aligned}\alpha_t(i)\beta_t(i) &= P(o_{1\dots t}, S_t = i)P(o_{t+1\dots T} \mid S_t = i) \\ &= P(o_{1\dots T}, S_t = i) \quad (\text{chain rule} + \text{cond. indep.}) \\ P(S_t = i \mid O) &\propto P(o_{1\dots T}, S_t = i) = \alpha_t(i)\beta_t(i)\end{aligned}$$

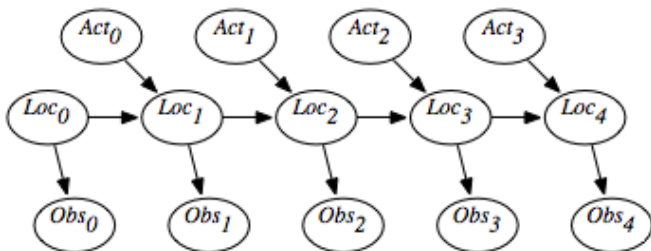
Dynamic Bayesian Networks (DBNs)

- In general, any Bayesian network can repeat over time: **DBN**
- Many examples can be solved with **variable elimination**,
- May become too complex with enough variables
- **Event-driven** times or **clock-driven** times



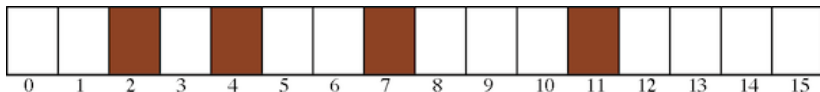
Example: Localization

- Suppose a robot wants to determine its location based on its actions and its sensor readings: **Localization**
- This can be represented by the **augmented HMM**:



Example Localization Domain

- **Circular** corridor, with 16 locations:



- Doors at positions: 2, 4, 7, 11.
- Noisy Sensors
- Stochastic Dynamics
- Robot starts at an unknown location and must determine where it is, known as the **kidnapped robot** problem.
- see handout `robotloc.pdf`

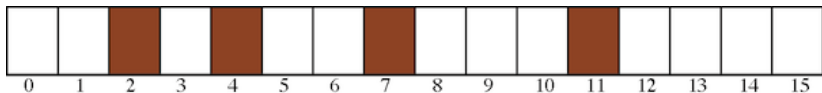
Example Sensor Model

- $P(\text{Observe Door} \mid \text{At Door}) = 0.8$
- $P(\text{Observe Door} \mid \text{Not At Door}) = 0.1$

Example Dynamics Model

- $P(\text{Loc}_{t+1} = \ell | \text{Action}_t = \text{goRight} \wedge \text{Loc}_t = \ell) = 0.1$
- $P(\text{Loc}_{t+1} = \ell + 1 | \text{Action}_t = \text{goRight} \wedge \text{Loc}_t = \ell) = 0.8$
- $P(\text{Loc}_{t+1} = \ell + 2 | \text{Action}_t = \text{goRight} \wedge \text{Loc}_t = \ell) = 0.074$
- $P(\text{Loc}_{t+1} = \ell' | \text{Action}_t = \text{goRight} \wedge \text{Loc}_t = \ell) = 0.002$ for any other location ℓ' .
 - All location arithmetic is modulo 16.
 - The action *goLeft* works the same but to the left.

Example sequence



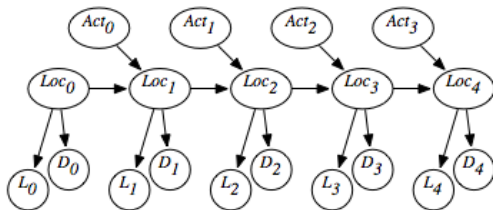
observe door, go right, observe no door, go right, observe door

Where is the robot?

$$P(\text{Loc}_2 = 4 \mid O_0 = d, A_0 = r, O_1 = \neg d, A_1 = r, O_2 = d) = 0.42$$

Combining Sensor Information

- **Example:** we can combine information from a light sensor and the door sensor through **sensor fusion**
- **Key point:** Bayesian probability ensures that evidence is integrated **proportionally to its precision**
- Sensors are **precision weighted**



Loc_t robot location at time t

D_t door sensor value at time t

L_t light sensor value at time t

Probability Distribution and Monte Carlo

- VE: exact algorithm
- Sometimes these calculations can be difficult,
- or the probability distributions are unknown
- Resort to stochastic sampling -
- So long as we can sample from a distribution, we can estimate it

Stochastic Simulation

- **Idea:** probabilities $\leftrightarrow$ samples
- Get probabilities from samples:

X	<i>count</i>
x_1	n_1
$\vdots$	$\vdots$
x_k	n_k
<i>total</i>	m

 $\leftrightarrow$

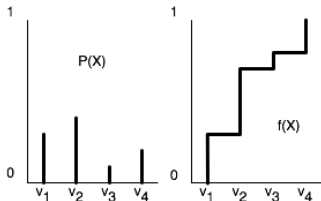
X	<i>probability</i>
x_1	n_1 / m
$\vdots$	$\vdots$
x_k	n_k / m

- If we could **sample** from a variable's (posterior) probability, we could **estimate** its (posterior) probability.

Generating Samples from a Distribution

For a variable X with a discrete domain or a (one-dimensional) real domain:

- **Totally order** the values of the domain of X
- Generate the **cumulative probability distribution**: $f(x) = P(X \leq x)$
- Select a value y **uniformly** in the range $[0, 1]$
- Select the x such that $f(x) = y$



Forward Sampling in a Belief Network

- Sample the variables **one at a time**
- Sample **parents** of X **before** you sample X
- Given values for the parents of X , sample from the **probability of X given its parents**
- For samples $s_i, i = 1 \dots N$:

$$P(X = x_i) \propto \sum_{s_i} \delta(x_i) = N_{X=x_i}$$

where

$$\delta(x_i) = \begin{cases} 1 & \text{if } X = x_i \text{ in } s_i \\ 0 & \text{otherwise} \end{cases}$$

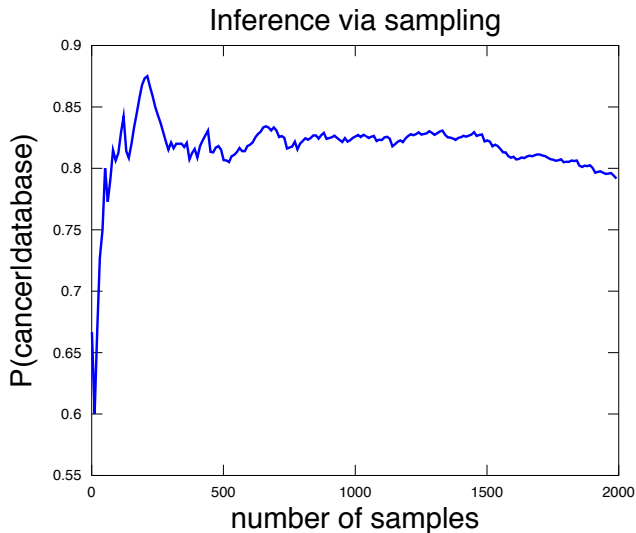
Sampling for a Belief Network: Inference

Sample	Malfnction	Cancer	TestB	TestA	Report	Database
s_1	false	false	true	true	false	false
s_2	false	true	true	true	true	true
s_3	false	true	true	true	true	true
s_4	false	false	false	true	false	false
s_5	true	true	true	true	false	false
s_6	false	true	false	true	false	false
s_7	false	false	false	true	false	true
...						
s_{1000}	false	false	false	true	false	false

To get $P(H = h_i | E = e_i)$ simply

- Count the number of samples that have $H = h_i$ and $E = e_i$, $N(h_i, e_i)$
- Divide by the number of samples that have $E = e_i$, $N(e_i)$
- $P(H = h_i | E = e_i) = \frac{P(H=h_i \wedge E=e_i)}{P(E=e_i)} = \frac{N(h_i, e_i)}{N(e_i)}$
- $P(C = \text{True} | \text{Database} = \text{True})$ based on first 7 samples?

Forward Sampling



Next

Bayesian Learning: Chapt. 10.1, 10.2, 10.4, 7.5